C CRYPTONITE

INSTITUTIONAL RELEASE V0.2.0

Quantitative Trading System User Manual & Operations Guide

A comprehensive operational manual for quantitative traders, portfolio risk officers, and infrastructure engineers. Features live application walkthroughs, Nautilus Trader backtesting, Polars vectorized scanners, and Gemini AI operations.

CLASSIFICATION	RELEASE VERSION
Institutional Trading Workstation	v0.2.0-STABLE
PRIMARY KERNEL	AI ENGINE
Nautilus Trader & Polars	Google Antigravity SDK & Gemini

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Generated Live: September 2026

1. Welcome & Quick Start Guide

Cryptonite is an institutional-grade, low-latency cryptocurrency algorithmic trading workstation. It unites tick/candle streaming, high-frequency technical screener calculations, discrete-event backtesting, and full operator trade governance within a modern single-pane dashboard.

1.1 Core Workstation Access Endpoints

Component / Endpoint	Network URL	Authentication	Description
Main Trading Terminal	`http://localhost:3000`	Public / Read-Only	Live candlestick charting, watchlists, and Polars scanner studio.
Operator Governance Portal	`http://localhost:3000/operator`	Operator Password	Hardware kill-switch, manual execution, strategy deployment.
FastAPI REST Gateway	`http://localhost:8000/docs`	Bearer / Cookie	Interactive Swagger API documentation and system endpoints.
AI Agent Operations	`http://localhost:8085`	Internal RPC / REST	Google Antigravity SDK quantitative agent fleet.

1.2 Default Operator Credentials

Administrative and trade execution features are guarded by an operator authentication layer:

- **Operator ID:** `admin`
- **Master Password:** `adminpass123`
- **Default Execution Environment:** `BINANCE DEMO / TESTNET` (Paper Capital)

> **Institutional Safety Default**
> The frontend starts in a strict **READ ONLY** state. Mutating actions (such as deploying strategies, changing leverage, or placing trades) require explicit operator authorization.

2. Trading Terminal & Live Market Data

The **Trading Terminal** serves as the central command cockpit for market surveillance, real-time telemetry, and technical analysis.

FIGURE 2.1 — CRYPTONITE TRADING TERMINAL & WATCHLIST LIVE WORKSPACE: TERMINAL

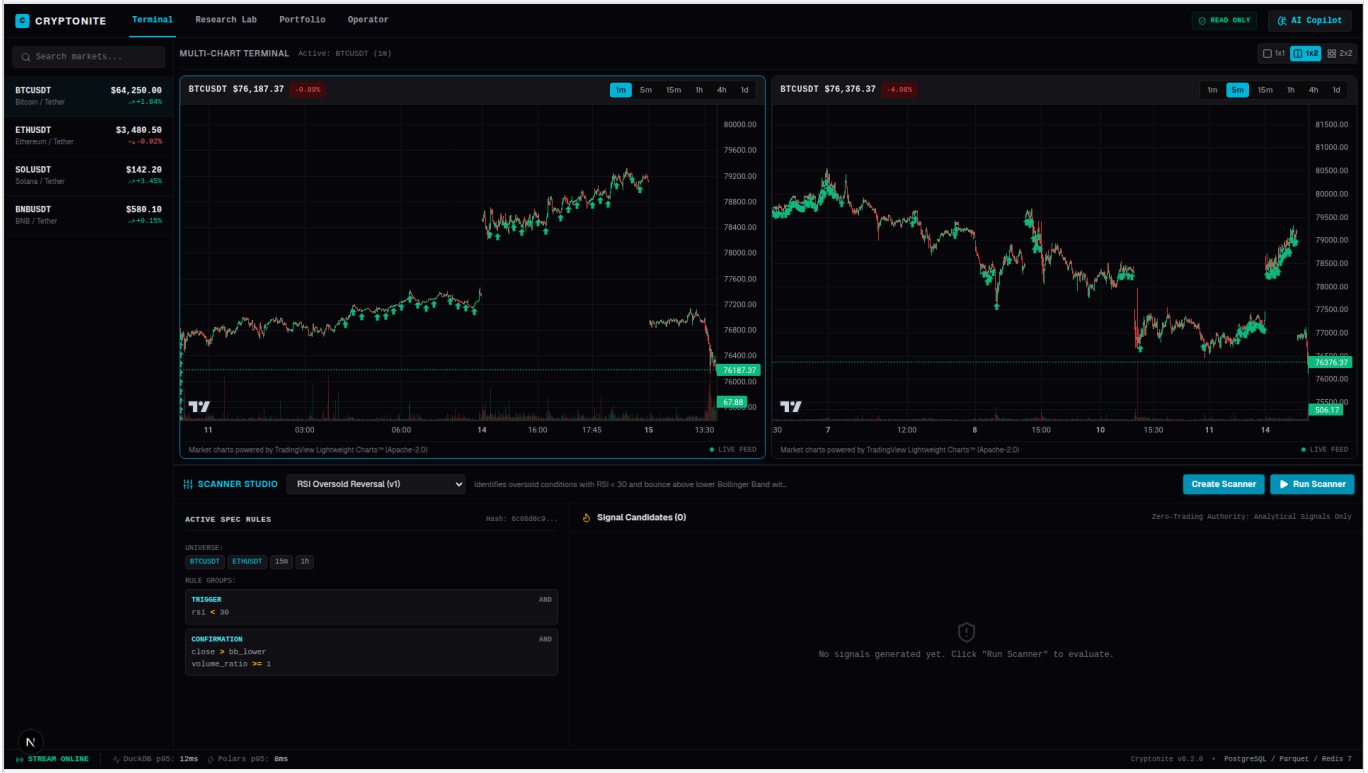


Figure 2.1: The primary trading terminal showing BTCUSDT candlestick price action on TradingView Lightweight Charts, the left-hand multi-asset watchlist, and the integrated Polars Scanner Studio.

2.1 Core Workspace Components

1. **TradingView Lightweight Charting:** Real-time hardware-accelerated canvas rendering live candlestick ticks directly from Binance WebSocket streams. Supports zooming, crosshairs, pan controls, and volume profile overlay.
2. **Timeframe Controls:** Select between `1m`, `5m`, `15m`, `1h`, `4h`, and `1d` intervals with zero-latency buffer switches.
3. **Multi-Asset Watchlist:** Displays 24-hour percentage change, bid/ask spread, and real-time prices across major pairs (`BTCUSDT`, `ETHUSDT`, `SOLUSDT`). Clicking any asset immediately updates the chart and scanner scope.
4. **Sub-Millisecond Engine Telemetry:** The footer displays live latency benchmarks:
   - `DuckDB p95: 12ms` (Columnar aggregation latency)
   - `Polars p95: 8ms` (Vectorized screening throughput)
   - `Stream: ONLINE` (Continuous WebSocket subscription status)

3. Vectorized Polars Scanner Studio

Cryptonite includes a vectorized scanner powered by **Polars LazyFrames** capable of calculating technical indicators and filtering thousands of candle bars across the market in single-digit milliseconds.

FIGURE 3.1 — VECTORIZED SCANNER BUILDER & STUDIO COMPONENT: ScannerBuilder.tsx

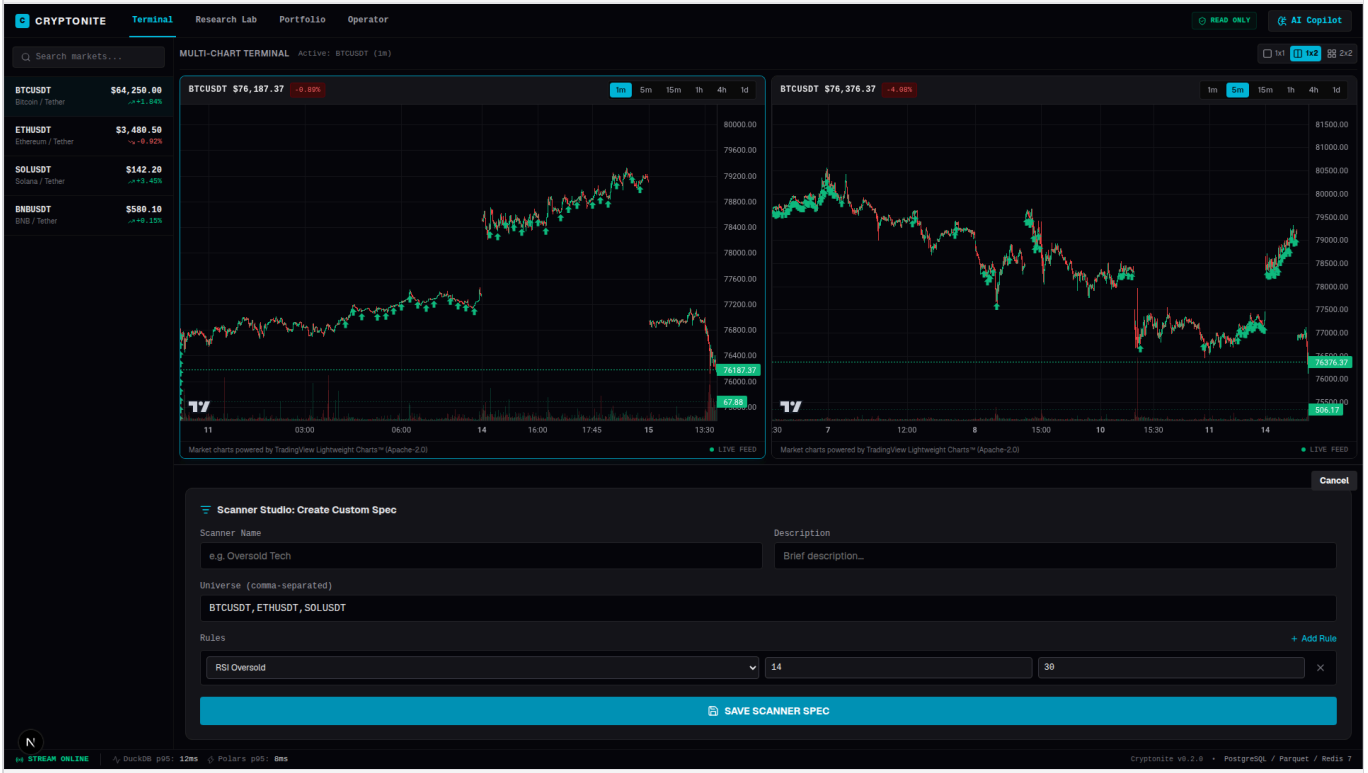


Figure 3.1: The interactive Scanner Builder modal enabling rapid visual composition of custom algorithmic filters with Polars expressions.

3.1 Creating and Running a Scanner

1. On the Scanner Studio header, click `CREATE SCANNER`.
2. **Specify Identifier & Target:**
   - Enter a memorable name (e.g., *RSI Oversold Momentum Reversal*).
   - Choose the primary instrument (`BTCUSDT.BINANCE`) and candle interval (`1m`, `15m`, `1h`).
3. **Configure Mathematical Conditions:**
   - Select the indicator: `RSI`, `Bollinger Bands`, `EMA Cross`, `Volume Spikes`, or `Supertrend`.
   - Set comparison operators (`<`, `>`, `CROSSES_ABOVE`, `CROSSES_BELOW`).
   - Define numeric thresholds (e.g., RSI Period: `14`, Upper Threshold: `70`, Lower Threshold: `30`).
4. Click **Save & Activate Scanner**.
5. Click `RUN SCANNER` to evaluate the condition against the Parquet catalog. All matching candles are displayed in the results table. Clicking any result row centers the main TradingView chart on that exact candle timestamp.

4. Quantitative Research Lab & Nautilus Backtesting

The **Research Lab** integrates **Nautilus Trader**—a state-of-the-art discrete-event algorithmic trading simulation kernel. It runs historical backtests that account for realistic network latency, slippage models, and exchange taker/maker fee structures.

FIGURE 4.1 — RESEARCH LAB & QUANTITATIVE TEARSHEET WORKSPACE: ResearchLab.tsx

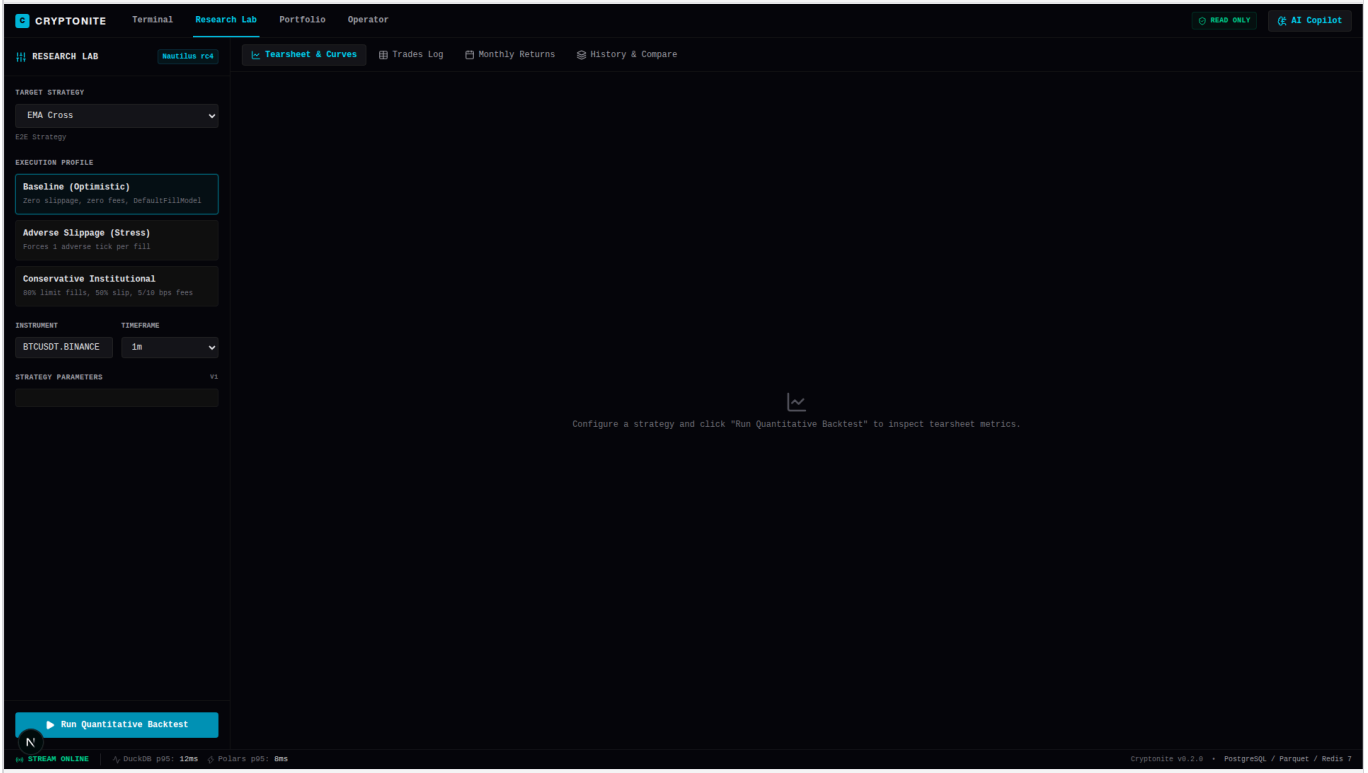


Figure 4.1: Research Lab interface with strategy selector, parameter customization, execution profile options, and comprehensive performance tearsheet visualization.

4.1 Running a Backtest Simulation

1. Navigate to the **Research Lab** tab on the top navigation bar.
2. **Strategy Selection:** Choose from available quantitative strategies (e.g., `EMA Cross`, `Breakout Volatility`).
3. **Hyperparameter Tuning:** Adjust numeric parameters (e.g., `Fast Period = 9`, `Slow Period = 21`).
4. **Execution Profile Selection:**

Execution Profile	Simulated Latency	Slippage Model	Taker / Maker Fee
`REALISTIC`	50ms	Volume-weighted tick slippage	1.0 bps
`PERFECT`	0ms	Zero slippage (mid-price fills)	0.0 bps
`HIGH VOLATILITY`	150ms	5.0 bps adverse drift	2.5 bps

5. Click `RUN QUANTITATIVE BACKTEST`. The backtester automatically verifies chronological data cadence, runs the simulation, and renders the tearsheet.

4.2 Interpreting Tearsheet Analytics

- **Sharpe & Sortino Ratios:** Measures risk-adjusted excess returns, penalizing downside volatility.
- **Max Drawdown (MDD):** Maximum peak-to-trough capital decline throughout the test duration.
- **Calmar Ratio:** Annualized Return divided by Maximum Drawdown.
- **Monthly Heatmap:** Color-coded matrix detailing month-over-month performance alpha.
- **Trades Ledger:** Full chronological log showing executed orders, timestamps, prices, and net PnL.

5. Portfolio Console & Risk Analytics

The **Portfolio Console** provides portfolio managers with unified capital accounting, real-time margin utilization tracking, and active perpetual futures position monitoring.

FIGURE 5.1 — PORTFOLIO CONSOLE & MARGIN METRICS WORKSPACE: PortfolioConsole.tsx

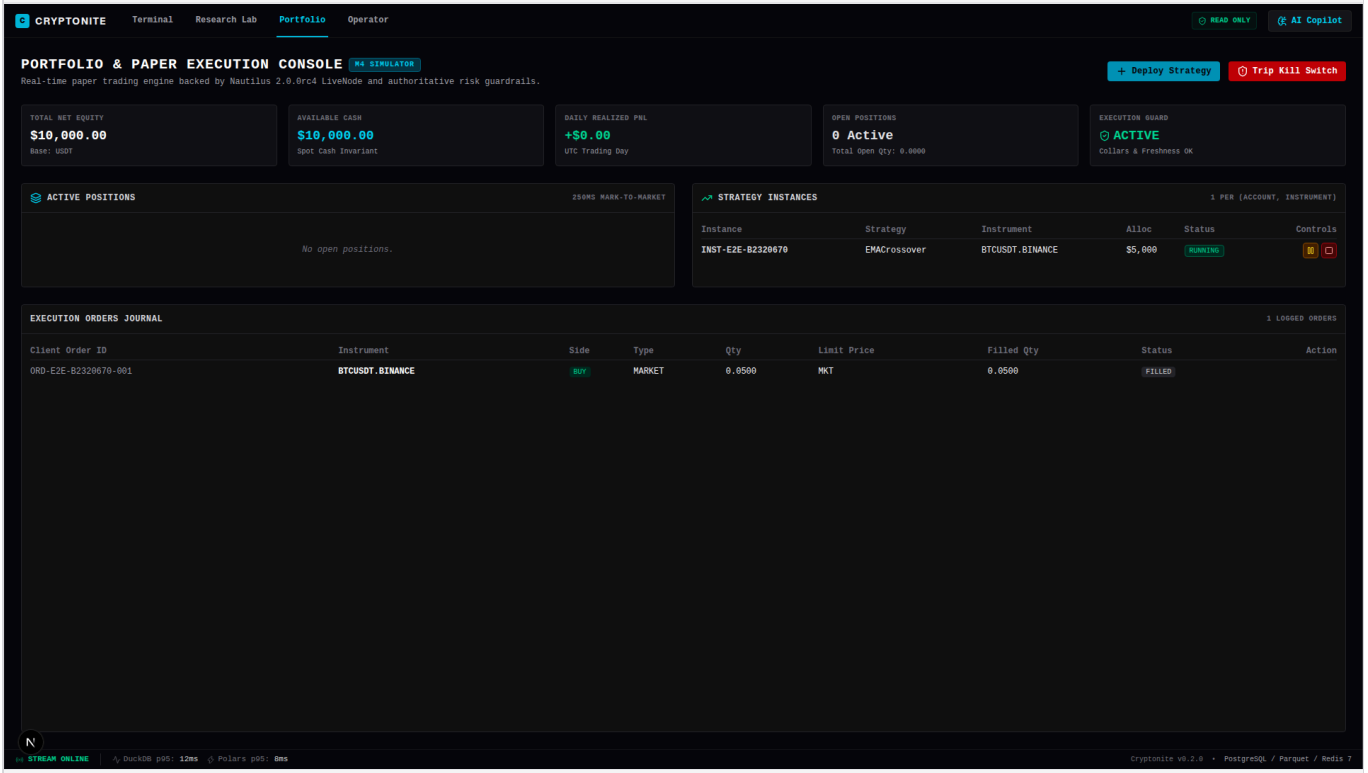


Figure 5.1: Real-time portfolio overview detailing total account collateral, margin ratios, position tables, and capital allocation across trading instruments.

5.1 Margin Risk Management

Cryptonite implements strict multi-tier margin risk models for perpetual contracts:

- **Margin Modes:** Supports both `CROSS MARGIN` (shared collateral pool) and `ISOLATED MARGIN` (risk segregated per position).
- **Initial Margin (IM):** Calculated as: $$IM = rac{ ext{Notional Value}}{ ext{Leverage}} = rac{Qty imes MarkPrice}{Leverage}$$
- **Maintenance Margin (MM):** Tiered formula reflecting Binance Futures risk brackets. If account equity drops below Maintenance Margin, the automated liquidation protocol triggers.
- **Liquidation Price Protection:** Displays real-time estimated liquidation distance per position to prevent catastrophic margin calls.

6. AI Copilot & Multi-Agent Operations

Cryptonite embeds an autonomous AI quantitative operations fleet powered by the **Google Antigravity SDK** and **Google Gemini** models. The copilot serves as an interactive partner for strategy research, performance debugging, and system telemetry analysis.

FIGURE 6.1 — AI COPILOT INTERACTIVE ASSISTANT AGENT: fleet.py (Antigravity SDK)

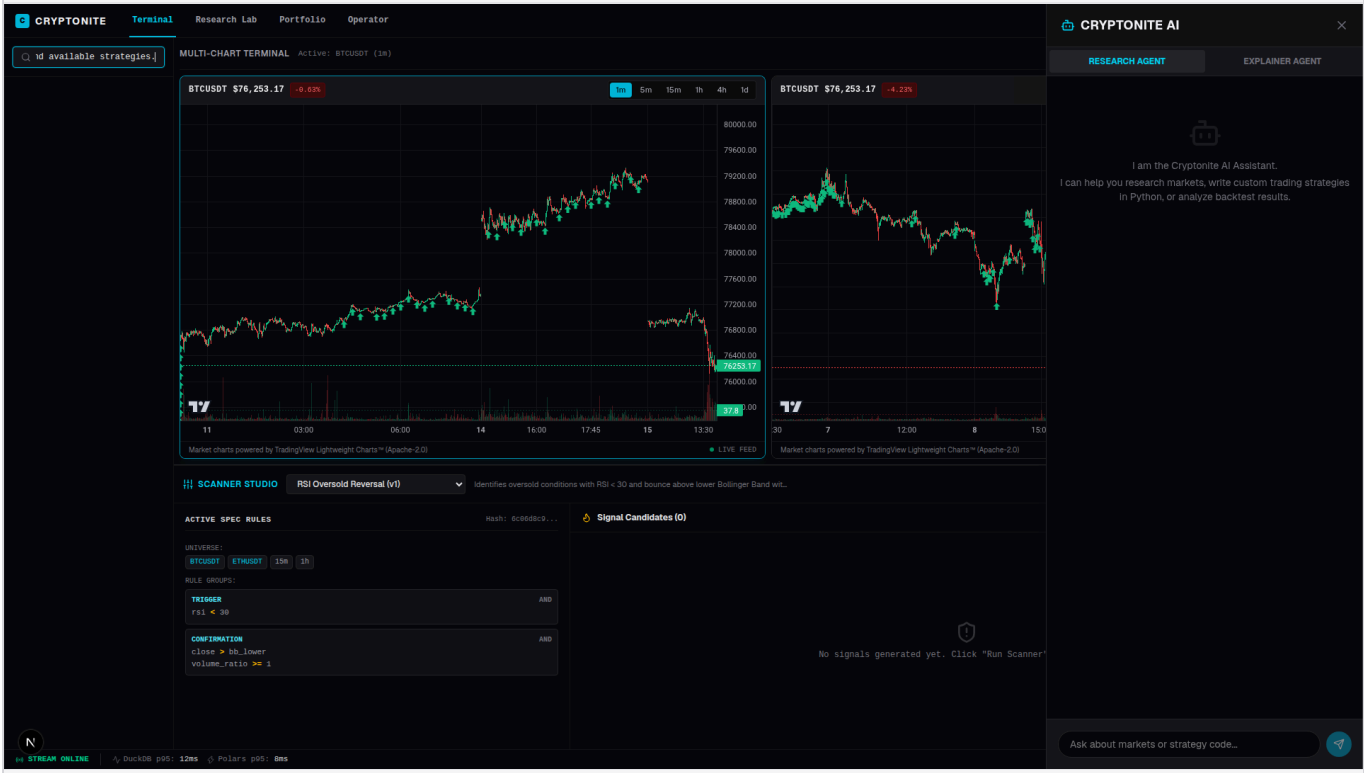


Figure 6.1: The AI Copilot drawer responding to operator queries regarding platform architecture, quantitative performance, and active trading strategies.

6.1 Specialized Agent Roles

1. **Research Agent:**
   - Directly queries historical market data Parquet catalogs using embedded **DuckDB** tools.
   - Performs sensitivity analysis on backtest runs to highlight overfitting or regime vulnerabilities.
2. **Explainer Agent:**
   - Accesses the PostgreSQL audit database via a safe, read-only unprivileged role (`cryptonite_agent_ro`).
   - Explains execution reports, order state machine transitions, and reconciliation ledger anomalies.

> **Enterprise Security Architecture**
> The AI Agent fleet is mathematically isolated from execution keys. Agents run in a dedicated container (`cryptonite_agent_worker`) using a read-only database transaction replica, preventing unintended order submissions or data modifications.

7. Operator Control Plane & Emergency Governance

The **Operator Control Plane** is designed for senior risk officers and operations engineers who hold operational responsibility over trading capital.

7.1 Operator Authentication

Accessing `/operator` requires entering valid operator credentials. Sessions are backed by secure, memory-hard Argon2id hashes and HTTP-Only session cookies.

FIGURE 7.1 — OPERATOR RESTRICTED ACCESS GATEWAY AUTH: /operator/login

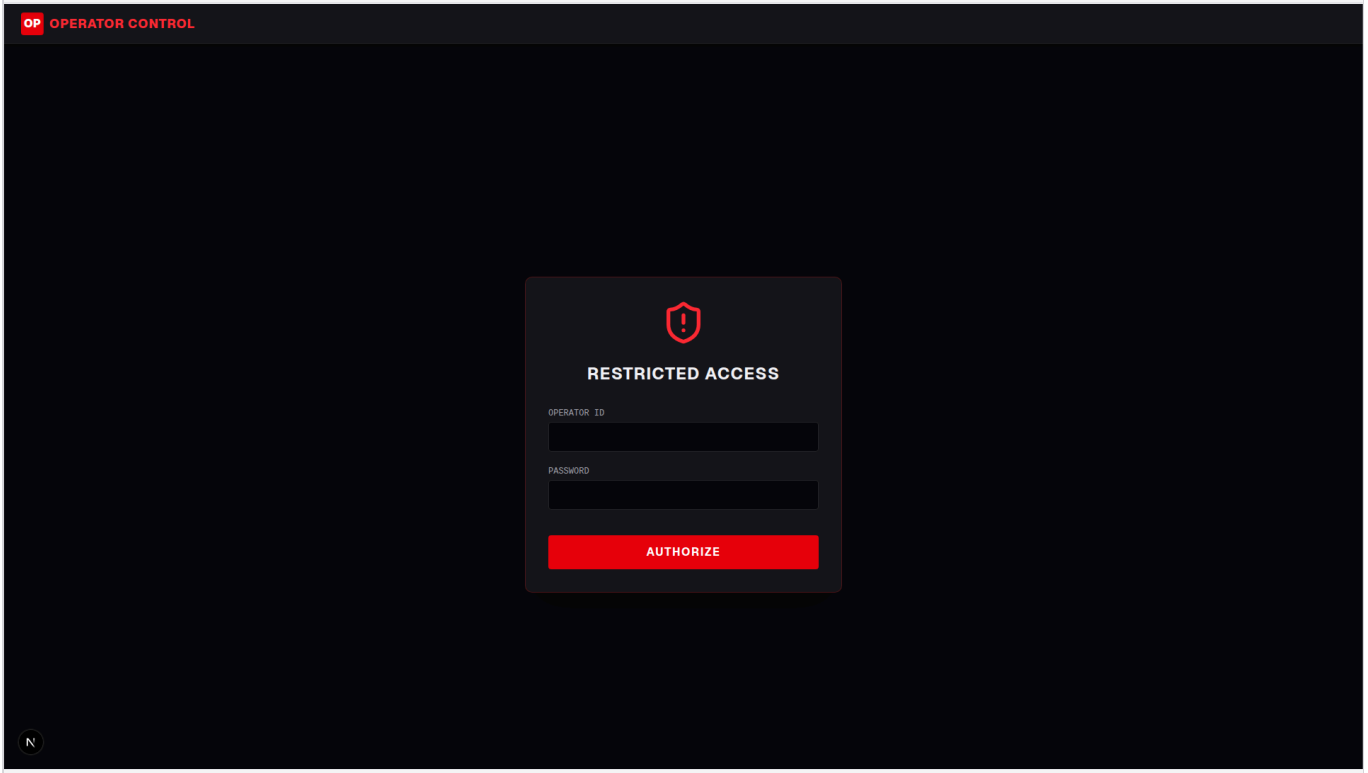


Figure 7.1: Operator restricted access login screen protecting administrative functions and live trading controls.

7.2 Operator Dashboard Overview

Upon login, the full operator console unlocks with execution controls, real-time risk gauges, strategy instance monitors, and reconciliation ledgers.

FIGURE 7.2 — OPERATOR GOVERNANCE DASHBOARD PANEL: OperatorDashboard.tsx

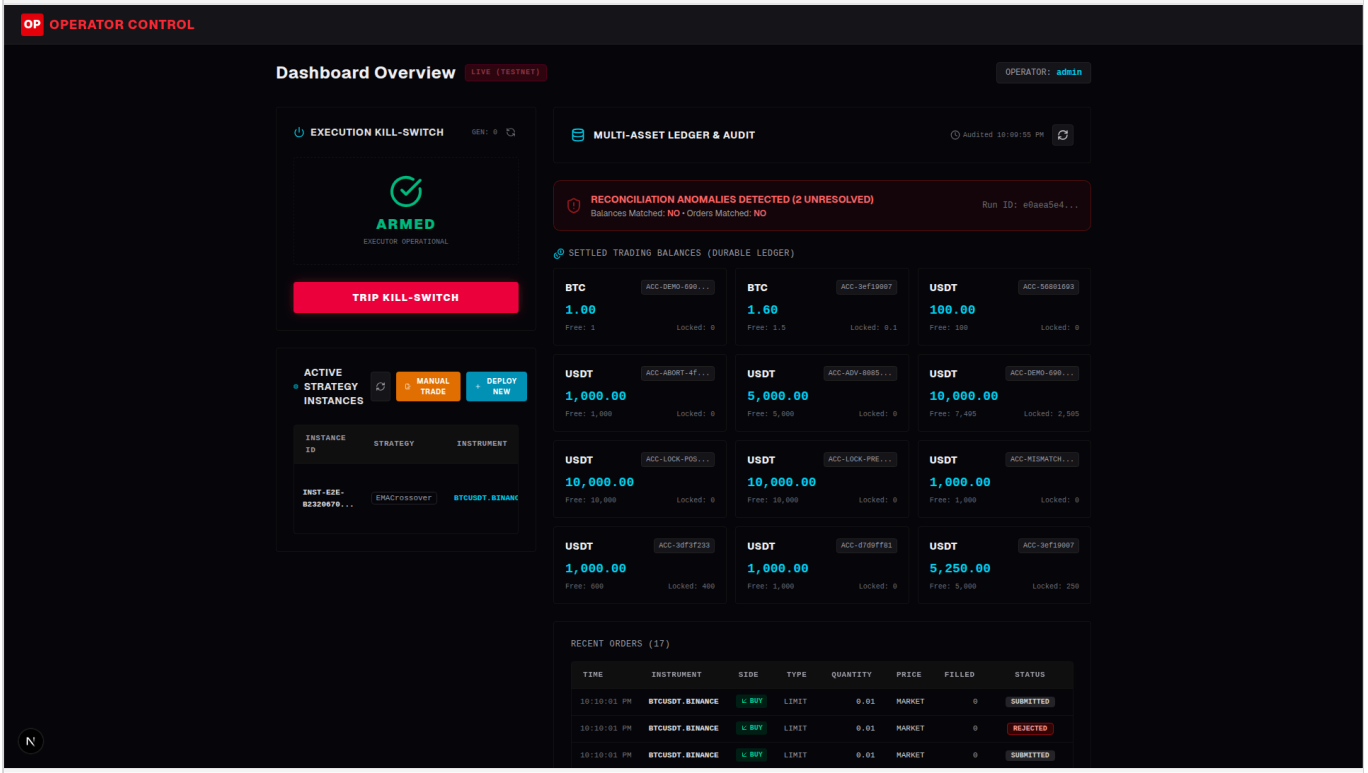


Figure 7.2: Comprehensive operator view showing the armed Execution Kill-Switch, active strategy instances (with live EMACrossover instance), and the Multi-Asset Reconciliation Ledger.

7.3 The Global Execution Kill-Switch

The top-left panel contains the system-wide emergency kill-switch:

- `ARMED STATE`: All deployed strategies submit orders normally.
- `TRIP KILL-SWITCH`: In the event of an exchange outage, runaway logic, or black swan event, clicking the red button initiates the shutdown sequence:
  1. Increments the atomic kill-switch generation counter in Redis and PostgreSQL.
  2. Instantly aborts all active strategy execution loops.
  3. Dispatches mass order cancellation requests to Binance for all open resting orders.
  4. Rejects any incoming orders with `HTTP 423 Locked`.
- **Re-Arming:** Disarming the kill-switch requires manual operator confirmation.

7.4 Deploying Strategy Instances

Click the `DEPLOY NEW` button on the Active Strategy Instances card to launch a strategy into live execution.

FIGURE 7.3 — DEPLOY STRATEGY INSTANCE MODAL — MODAL: DeployStrategyModal.tsx

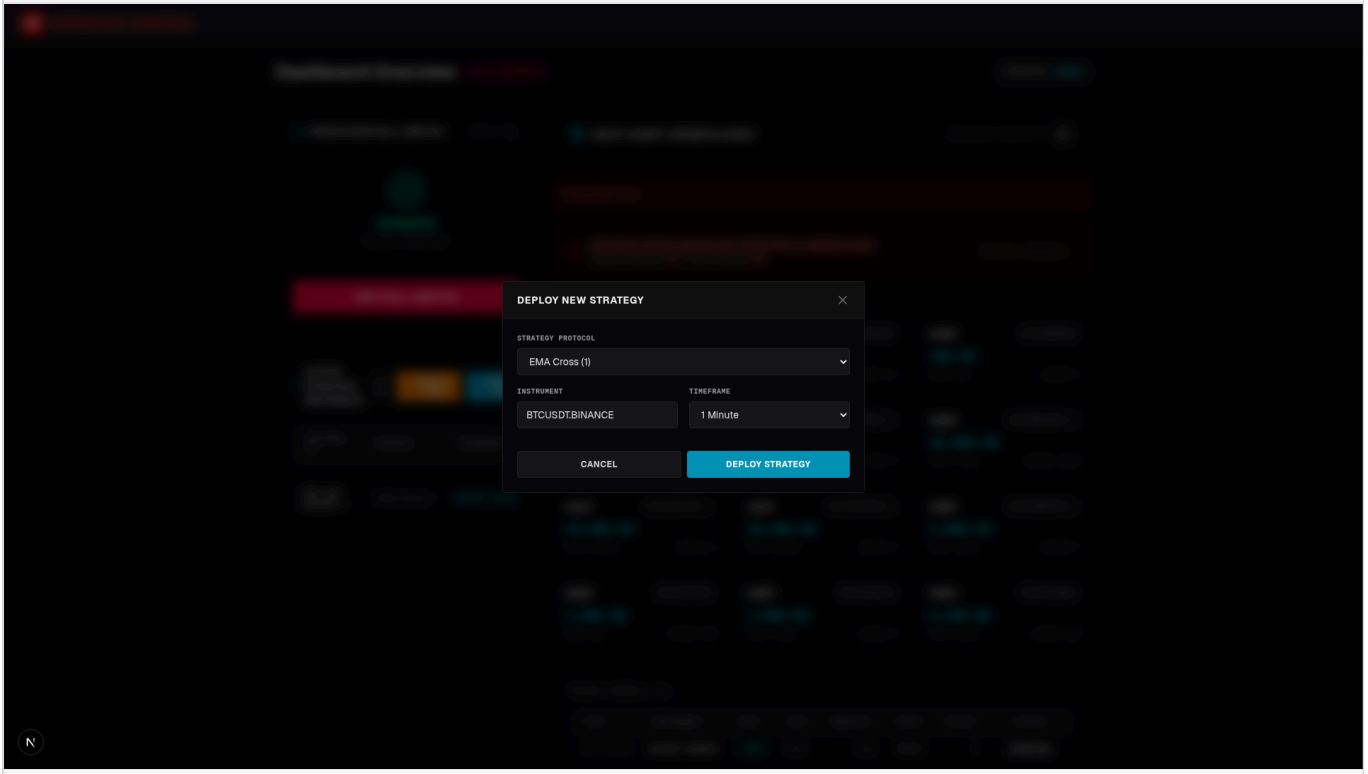


Figure 7.3: Deploy Strategy Modal allowing operators to select strategy definitions, target instruments, capital allocation, and leverage limits.

7.5 Executing Manual Emergency Trades

Click the `MANUAL TRADE` button to directly dispatch OTC orders or manually rebalance open inventory.

FIGURE 7.4 — MANUAL TRADE DISPATCH MODAL MODAL: ManualTradeModal.tsx

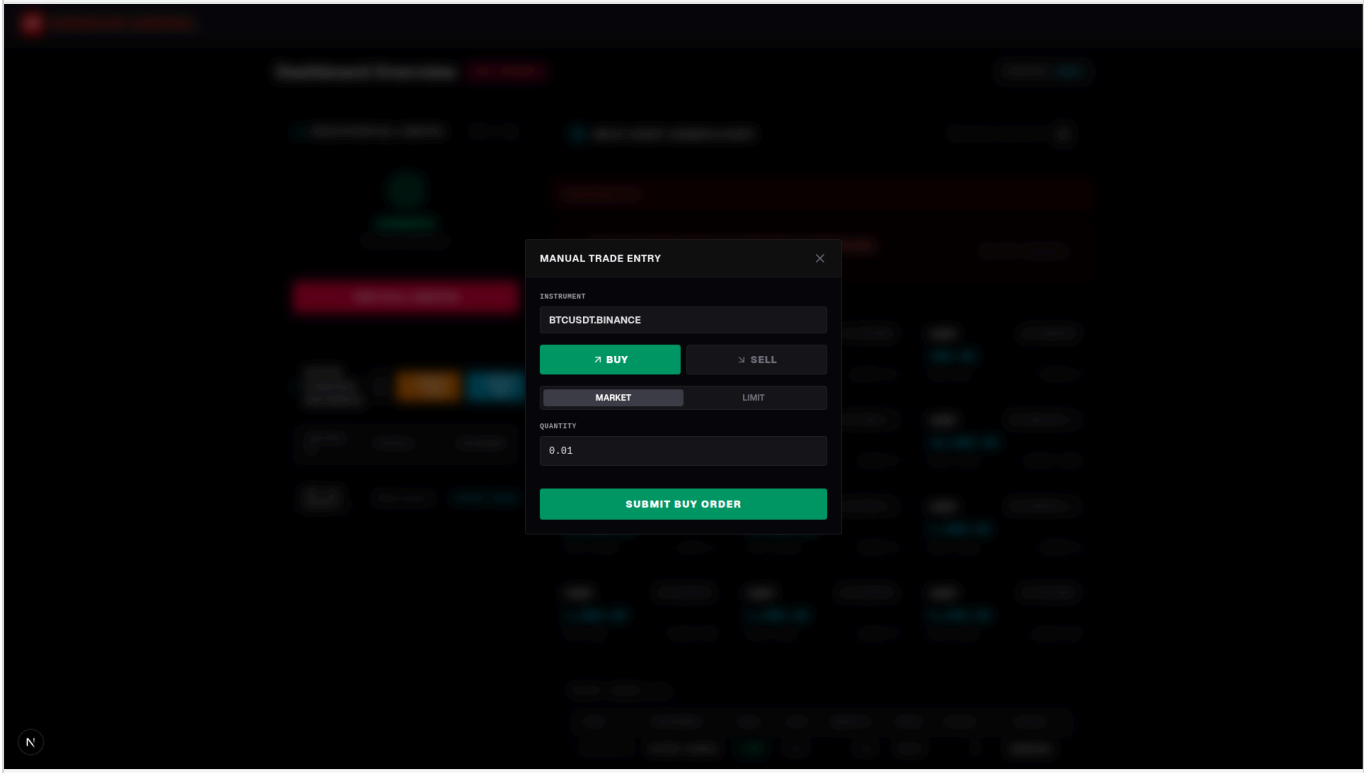


Figure 7.4: Manual Trade modal supporting direct market or limit order dispatch with instantaneous notional calculations.

7.6 Multi-Asset Reconciliation Ledger

Cryptonite continuously audits internal database state against live Binance exchange balances and open orders.

- **Audited Balances:** Validates settled balances across all quote and base currencies.
- **Order State Alignment:** Detects phantom fills, orphan orders, or lost network acknowledgments.
- **Anomaly Alerting:** Any delta is immediately flagged in amber/red with an audit run ID for post-mortem forensics.

8. Troubleshooting & Frequently Asked Questions

8.1 Frequently Encountered Issues

AI Copilot: "Gemini API key is required"

Resolution: Ensure your `GEMINI_API_KEY` is defined in the root `.env` file. The `infra/docker-compose.yml` file automatically injects this into the `agent_worker` container. If modified, restart the container:

```
cd infra && docker compose up -d agent_worker
```

Backtester: "Out of order or duplicate bars detected"

Resolution: Nautilus Trader strictly requires historical data bars to be in perfect chronological order without duplicates. Cryptonite's cadence validator automatically sorts and deduplicates timestamps before starting backtests. If you ingest raw parquet files manually, ensure timestamps are strictly monotonic.

WebSocket Feed: "CONNECTING..."

Resolution: Verify that the `cryptonite_ingest` and `cryptonite_redis` containers are healthy. The web client connects to the Redis stream via the FastAPI bridge on port `8000`. Check connection status via:

```
curl http://localhost:8000/health
```

8.2 Container Administration Cheat Sheet

Administrative Action	CLI Command
Check container health & status	`cd infra && docker compose ps`
Tail live logs of a specific service	`cd infra && docker compose logs -f <service_name>`
Rebuild and restart a modified service	`cd infra && docker compose build <svc> && docker compose up -d <svc>`
Inspect running database migrations	`docker exec cryptonite_api alembic current`
Trigger manual reconciliation audit	`curl -X POST http://localhost:8000/api/v1/control/reconcile`

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